Lie Detection Internship 2025

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Contents

1 Introduction

This report summarizes the work conducted during the internship on EEG-based lie detection. Several objectives were set for the internship:

- Review recent literature on the subject.
- Expand the existing NTNU EEG dataset.
- Apply methods described in the literature to the dataset.

An intern (Louis Ossant) had already worked on this subject last year. He designed the protocol for EEG measurements and attempted to identify the most relevant EEG channels for lie detection. His work is available in his report on GitHub.

2 State of the Art

Recent advancements in EEG-based lie detection have demonstrated significant progress through diverse methodological approaches. Researchers have introduced new datasets such as LieWave [1], which provides a comprehensive framework for evaluating deception detection algorithms. While traditional feature extraction methods, such as those proposed in [2], laid the groundwork for analyzing EEG signals, more recent techniques such as wavelet-based feature extraction [3] have shown potential, though with modest accuracy (72% using Naïve Bayes). A notable result was reported in [4], achieving exceptional performance (99.96% accuracy) using advanced classification techniques. A review by [5] synthesizes the state of the art, covering devices, feature extractors, and classifiers, and highlighting key trends in the field. Additionally, novel algorithms such as ATAR [6] and CSP L1 [7] have been proposed, offering improved signal processing. These contributions underline the rapid evolution of EEG-based lie detection in recent years, with increasing accuracy and robustness.

3 LieWave Dataset

3.1 Preprocessing and Feature Extraction

The LieWave Dataset was developed in [1]. We attempted to replicate some of the methods described in that work. The dataset consists of EEG signals from 27 subjects, each performing two runs (one truth-telling and one lying). The original data dimensions were structured into 54 runs, 25 stimuli per test, 5 channels, and 384 samples per channel. The dataset includes four preprocessing variants:

- Band-pass filter (0.5–45 Hz)
- Artifact Subspace Reconstruction (ASR)
- Independent Component Analysis (ICA)
- ATAR (Automatic and Tunable Artifact Removal)

We tested different feature extraction methods inspired by [2], applying them to each preprocessed signal. The most relevant were:

- Fast Fourier Transform (FFT)
- Power Spectral Density (PSD)
- Discrete Wavelet Transform (DWT)

For each transformed signal, we extracted statistical features (mean, variance, skewness, etc.).

Table 1: Final feature dimensions of the LieWave dataset before classification.

Method	Dimensions
Pre-Processed	(1350, 1920)
FFT	(1350, 30)
PSD	(1350, 150)
DWT	(1350, 100)

3.2 Classification

Five classifiers were evaluated:

- Support Vector Machine (SVM)
- Random Forest
- k-Nearest Neighbors (KNN)
- XGBoost
- Logistic Regression

Table 2 summarizes the top six results for the LieWave dataset. The best performance was achieved with ASR preprocessing and FFT features using XGBoost (82% accuracy).

Table 2: Top six classification results for the LieWave dataset (accuracy).

Preprocessing	Features	Classifier	Accuracy
ASR	FFT	XGBoost	0.82
ASR	PSD	Random Forest	0.81
ASR	PSD	XGBoost	0.80
ASR	DWT	Random Forest	0.78
ASR	DWT	XGBoost	0.77
Band-pass	DWT	Random Forest	0.76

With the LieWave dataset, we obtained consistent results across different feature extraction methods and classifiers. We did not achieve the same performance reported in the original paper, as they used more advanced methods. Nevertheless, working on this dataset helped us become familiar with EEG signals, preprocessing techniques, and feature extraction, which will be useful when analyzing the NTNU dataset.

4 NTNU Dataset

The NTNU dataset was originally developed by the previous intern, and we expanded it by collecting new measurements.

4.1 Previous Work

The NTNU dataset consisted of 568 test runs with 32 channels and 251 samples per channel. The previous intern's work describes the measurement protocol in detail.

4.2 Our Work

We encountered issues when running some tests with the EEG device, so we modified Louis's protocol slightly. Most changes were made in the Controller file to capture data and transform it into epochs. As we did not have access to the images he used, we created code to generate our own stimuli, which is now available for future use. Figure 1 shows the lie detection protocol: participants choose whether to lie or tell the truth by clicking on an image.

In total, we collected 179 new test runs: 97 truth and 82 lie, with the same dimensions as the original dataset.

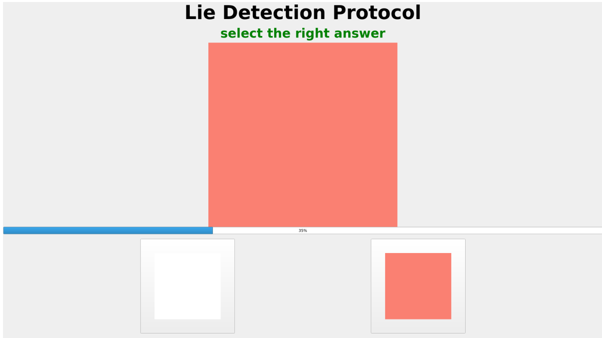

Figure 1: Lie detection protocol.

4.3 Choice of Channels

The previous intern concluded that channel selection is important, as not all channels are equally informative for lie detection. We used his identified "channels of interest" in our experiments.

To verify his findings, we generated PSD topographic maps. Figure 2 shows the map for Subject 1.

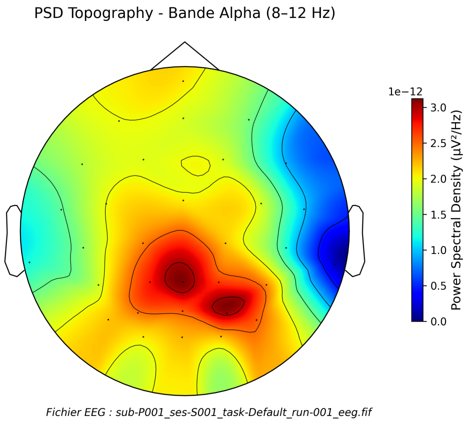

Figure 2: PSD topographic map for Subject 1.

The most active region (highest PSD, shown in dark red) is located in the posterior part of the scalp, likely within the occipital and parieto-occipital areas, typical for alpha activity. This activity is common at rest, especially with eyes closed, and indicates relaxation or attentional disengagement. The previously identified channels also appear more active on the topographic map, confirming their relevance.

4.4 Preprocessing

We tested several preprocessing methods for the NTNU dataset. First, we applied a band-pass filter (0.5–45 Hz). We then implemented the ATAR algorithm [6], but it yielded poor results (around 50% accuracy). Based on advice, we applied the CSP algorithm, specifically the L1-norm variant described in [7], which improved performance. We also tested PCA due to the dataset's dimensionality, which proved useful in some cases.

4.5 Feature Extraction and Classification

We used the same five classifiers as for the LieWave dataset. Given the lower performance, we tested many combinations of channels, preprocessing, and feature extraction. The best results are shown in Table 3.

Here, **CI** means "Channels of Interest" (the most relevant channels), while **AC** means "All Channels" (all 32 EEG channels).

The results are lower than for the LieWave dataset, which we attribute to:

Table 3: Classification accuracy for the NTNU dataset.

	SVM	Random Forest	KNN	XGBoost	Logistic Regression
CI ATAR	0.5088	0.4806	0.5053	0.5018	0.5256
CI CSP	0.6127	0.5423	0.5493	0.5282	0.6021
AC CSP	0.5855	0.5669	0.5370	0.5880	0.5898
CI CSP PCA99	0.5933	0.6000	0.5475	0.6039	0.6021

- Differences in EEG device quality, with LieWave likely using a higher-end system.
- A protocol that may not be strongly related to deception, but rather simple "yes/no" questions.
- Simpler feature extraction and classification methods compared to those used in the literature.

5 Conclusion

This internship provided valuable insight into EEG-based lie detection, from literature review to hands-on experimentation with two different datasets. Using the publicly available LieWave dataset, we reproduced and adapted published methodologies, achieving accuracies up to 82% with relatively simple feature extraction and classification techniques. Although our results did not match the highest reported in the literature, this work allowed us to strengthen our understanding of preprocessing pipelines, feature extraction, and classifier selection for EEG analysis.

On the NTNU dataset, we faced several challenges: technical limitations of the EEG device, protocol constraints, and differences in data quality. Despite experimenting with various preprocessing methods, channel selections, and dimensionality reduction techniques, classification accuracies remained modest (best results around 60%). Nevertheless, the work confirmed the relevance of certain channels for deception detection and validated the use of CSP-based preprocessing in our context.

Here are the future possibilities to improve the NTNU Dataset :

- Working on a better protocol, that underline the deception
- Continue to search in the literature for better methods

6 Advice for Future Interns

As this report will be on GitHub, future interns might read it. Here is some advice to avoid common mistakes:

- Understanding `mne` Epoch objects is crucial, as they are the core EEG data structure in Python.
- Epochs contain 33 channels: 32 EEG channels and 1 `stim` or `STI` channel, which encodes the labels (truth or lie) for each run. Remove the `stim` channel before processing EEG data to avoid data leakage. We overlooked this initially and lost time.
- The papers [5] and [1] are excellent starting points for understanding the field.

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